

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 312 — Past Questions & Materials
by Renance

Course: MTH 312
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — MTH 312 Ordinary Differential Equations 11 2020 2021 First Semester

Level 300 • First Semester • PastQuestion

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MARKING GUIDE
2020/2021

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
(FUPRE)

DEPARTMENT OF MATHEMATICS
2020/2021 First Semester Examination
MTH 312 - ORDINARY DIFFERENTIAL EQUATIONS II

Instruction: Answer ANY FOUR Questions

Time Allowed: $2\frac{1}{4}$ Hours

Date: 18th November, 2021

- (1) (a) What is a differential equation?
Any equation relating an unknown function with its derivatives is called Differential equations. i.e.

$$F(x, y, y', y'', \dots, y^{(n)}) = 0$$

3Marks

- (b) When is a differential equation said to be

(i) Ordinary
It is said to be ordinary if the dependent variable is depending only on one independent variable. [2]Marks

(ii) Partial
If the dependent variable is depending on more than one independent variables [2] Marks

(iii) Initial Value Problem (IVP)
This is a differential equation with associated condition(s) specified at an initial point called the initial conditions [2]Marks

(iv) Boundary Value Problem
This is a differential equation with associated condition(s) specified at point(s) on the interval representing the boundary called the boundary conditions [2]Marks

- (c) What is a solution of a differential equation?
Any function $\phi(x)$ that satisfies a differential equation on the given interval is called the solution of the differential equation. [3]Marks

- (d) Find the general solution to the problem $y'' + 4y' + 3y = 0$
The equation is a second order linear, homogeneous with constant coefficients differential equation. [1]Mark

The equivalent characteristic equation to the problem is

$$r^2 + 4r + 3 = 0$$

[1]Mark

which on factorising yields

$$(r + 1)(r + 3)$$

The characteristic roots are $r_1 = -1$ and $r_2 = -3$

[2]Marks

Therefore, the general solution of the problem is given as

$$y(x) = Ae^{r_1 x} + Be^{r_2 x}$$

[1]Mark

which is

$$y(x) = Ae^{-x} + Be^{-3x}$$

where A and B are arbitrary constants.

[1]Mark

Paper 2 of 2 — MTH 312, 2021

Level 300 • First Semester • PastQuestion

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(b) Evaluate the integral below:

$$\int_c \frac{(3z+5)e^{2z} dz}{2z-5}, c = \{z : |z| = 1\}; c = \{z : |z| = 3\}$$

(5 marks)

(c) Using (3a) above or otherwise, evaluate the following integrals:

(i) $\int_c \frac{z dz}{2z-1}, c = \{z : |z| = 1\}$

(ii) $\int_c \frac{3z dz}{(z^2-16)(z-4)}, c = \{z : |z| = 3\}; c = \{z : |z-4| = 1\}$ (6 marks)

Question 4:

(a) Suppose f is analytic in a certain neighborhood N of z_0 . If $|f(z)| \leq |f(z_0)|$ for all $z \in N$, then prove that $f(z)$ is constant on N . (6 marks).

(b) State the maximum modulus principle. Given that $f(z)$ is analytic on $\{z : |z| \leq 1\}$, $f(0) = 0$, and that for some $M > 0$, $|f(z)| \leq M$ on $\{z : |z| = 1\}$ by considering

$$g(z) = \frac{f(z)}{2M - f(z)}$$

or otherwise, show that

$$|f(z)| \leq \frac{2M|z|}{1+|z|}$$

for $|z| < 1$ (8 marks)

(c) Hence when does $f(z) = \sin z$ have a maximum? ($3\frac{1}{2}$ marks)

Question 5:

(a) State Roche's theorem. (3 marks).

(b) Prove that every non-constant polynomial has at least one zero ($8\frac{1}{2}$ marks)

(c) Given a polynomial of degree 5 with complex coefficients. How many possible number of roots can it have? Hence locate the zero of the polynomial

$$z^4 + z^3 + 1$$

on $c = \{z : |z|, 3\}$. (6 marks)

Question 6:

(a) Consider the family of polynomials

$$p(z, t) = a_n z^n + a_{n-1}(t) z^{n-1} + \dots + a_1(t) z + a_0(t)$$

where the coefficients $a_j(t)$ depend continuously on the parameter $t \in [0, 1]$. Assume that at $t = 0$, the polynomial $p(z, 0)$ has k zeroes (counted with their multiplicity) in the disk $|z - c| < \mathbb{R}$ and has no zeroes on the circle $|z - c| = \mathbb{R}$. Show that for all sufficiently small t the polynomial $p(z : t)$ also has k zeroes in $|z - c| \leq \mathbb{R}$. ($7\frac{1}{2}$ marks).

(b) Verify Roche's theorem with the polynomial $z^3 + 12z + 1$ inside the disk $|z| = 3$. (4 marks)

(c) Determine the number of zeroes for polynomials (i) $z^7 - 2z^3 - 7$ inside the disk $|z| = 2$ and (ii) $z^5 - 9z^2 + 11$ inside the disk $|z| = 4$. (6 marks).

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2020/2021

Course Code/ Title: MTH 313-Complex Analysis II
Instructions: Attempt any 4 questions. Time Allowed: $2\frac{1}{2}$ Hours

Question 1:

- (a) What is a singular point? When is a singular point said to be (i) isolated (ii) removable (iii) a pole and (iv) essential? (6 marks).
- (b) Let $\frac{z^3-1}{z^2+3z-4}$ be a complex-valued rational function with a power series expansion in a neighborhood of the origin. Find its singular points. Identify what type of singular points are they. (4 marks).
- (c) State Laurent's theorem. Find the Laurent series expansion about the indicated singularities for the functions below. Classify the singularities in each case and state the region of convergence of each series:
(i) $\frac{e^{2z}}{(z-1)^3}$, $z = 1$ (ii) $(z-3) \sin(\frac{1}{z+2})$, $z = -2$ (iii) $\frac{1}{z^2(z-3)^2}$, $z = 3$ ($7\frac{1}{2}$ marks).

Question 2:

- (a) Differentiate between (i) a single pole (ii) a double pole and (iii) a triple pole (3 marks).
- (b) State the Cauchy's residue theorem. Find the residue of the following:
(i) $f(z) = \frac{z^2-2z}{(z+1)^2(z^2+4)}$
(ii) $f(z) = \frac{e^{iz}}{z^2(z^2+2z+2)}$ (8 marks).
- (c) Using the Cauchy's residue theorem evaluate the following integrals

- (i) $\int_C \frac{ze^z dz}{z^2-1}$, $C = \{z : |z| = \frac{3}{2}\}$
(ii) $\int_C \frac{\sinh z dz}{z^3}$, where C is a circle with center at the origin. ($6\frac{1}{2}$ marks).

Question 3:

- (a) State the Cauchy's integral theorem. Let $f(z)$ be analytic within and on a simple closed contour C . Then prove that for any point z_0 in the interior of C :

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z) dz}{z - z_0}$$

($6\frac{1}{2}$ marks).

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